JSS1 DIGITAL TECHNOLOGIES LESSON NOTES (SECOND TERM)

WEEK 1: Introduction to MS PowerPoint

WHAT IS MICROSOFT POWERPOINT?

Microsoft PowerPoint is a computer program used to create **presentations**. A presentation is a collection of slides that can show text, pictures, and videos to an audience. It helps you share information in a clear and organized way.

EXAMPLE OF OTHER PRESENTATION SOFTWARE: Google Slides, Apple Keynote, Canva.

USES OF POWERPOINT:

- **In School:** Teachers use it to teach lessons; students use it to present projects.
- **At Work:** For business meetings and reports.
- **At Home:** To create photo slideshows for family events.
- **In the Community:** For announcements in churches or clubs.

THE POWERPOINT WINDOW (PARTS)

When you open PowerPoint, you'll see these main parts:

1. **Title Bar:** The bar at the very top that shows the name of your presentation.
2. **Ribbon:** The strip of tabs (like Home, Insert, Design) at the top. It contains all the tools you need.

3. **Slide Panel:** The area on the left that shows small pictures of all your slides.
4. **Slide Area:** The big, white space in the center where you design and work on your current slide.
5. **Notes Section:** The area at the bottom where you can write notes for yourself (the speaker) that the audience won't see.

CREATING AND SAVING SLIDES

- **Opening PowerPoint:** Click the Start menu, find Microsoft PowerPoint, and open it. A blank presentation usually starts automatically.
- **Creating a New Presentation:** Go to `File` → `New` → `Blank Presentation`.
- **Adding a New Slide:** Go to the `Home` tab and click `New Slide` (or press `Ctrl + M` on your keyboard).
- **Saving Your Work:**
  - **First time:** Click `File` → `Save As` → choose where to save it (like your Documents folder) → type a name for your file → click `Save`.
  - **After that:** Press `Ctrl + S` to save your changes quickly.

ADDING TEXT, IMAGES, AND EFFECTS

- **Adding Text:** Click on a text box that says "Click to add title" and start typing. Use the `Home` tab to change the font, size, and color.
- **Adding Images:** Go to the `Insert` tab → click `Pictures` → choose "This Device" to find a picture on your computer → select it and click `Insert`. You can drag the corners to resize it.
- **Moving Slides:** In the left slide panel, simply click and drag a slide up or down to change its order.
- **Deleting a Slide:** Click the slide you want to remove in the left panel and press the `Delete` key on your keyboard.

TRANSITIONS AND ANIMATIONS

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- **Transitions:** These are special effects that happen when you **move from one slide to the next**. (e.g., Fade, Push, Wipe).
  - *How to add:* Select a slide → go to the `Transitions` tab → choose an effect.
- **Animations:** These are special effects that happen to **objects on a single slide**, like text or a picture. (e.g., Appear, Fly In, Fade).
  - *How to add:* Select an object (like a text box) → go to the `Animations` tab → choose an effect.

HOW TO RUN YOUR PRESENTATION SLIDESHOW

- To start your presentation, go to the `Slide Show` tab and click `From Beginning`, or simply press the `F5` key on your keyboard.
- Click your mouse or press the spacebar to move to the next slide.
- Press the `Esc` key to exit the slideshow.

HOW TO WRITE A GOOD PRESENTATION:

1. Keep your text short and simple.
2. Use a large font size (at least 24) so everyone can read it.
3. Use clear colors that are easy on the eyes.
4. Don't put too much information on one slide.
5. Use pictures that relate to what you're saying.

Evaluation Questions:

1. What is Microsoft PowerPoint used for?
2. How do you add a new slide?
3. Explain the difference between a transition and an animation.
4. Name any three parts of the PowerPoint window.
5. How do you save a presentation for the first time?

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WEEK 2: 3D Modelling with Paint 3D

Topic: 3D Modelling with Paint 3D

Subtopic 1: Introduction to Paint 3D

- **What is it?** Paint 3D is a free app from Microsoft that you can use to create not just flat drawings, but also three-dimensional (3D) objects.
- **2D vs. 3D:**
  - **2D** objects are flat, like a drawing on a piece of paper. They have height and width.
  - **3D** objects have depth, like a real ball or a box. You can rotate them and look at them from all sides.
- **Opening Paint 3D:** Click the Start menu, type "Paint 3D", and open the app.

Subtopic 2: Drawing Basic Shapes and Models

- **The Workspace:** The big area in the middle is your **canvas**.
- **2D Shapes:** Click the "2D shapes" icon on the left. Choose a shape (like a circle or square), then click and drag on the canvas to draw it.
- **3D Shapes:** Click the "3D shapes" icon. Choose a shape (like a cube, sphere, or cone). It will appear on your canvas.
- **Working with 3D Objects:**
  - **Move:** Click and drag the object to move it.
  - **Resize:** Drag the white boxes on the corners of the object.
  - **Rotate:** Click and drag the circular handle at the top of the object to spin it around.
  - **Color:** Select the object, then click a color on the right panel.

Simple Model Ideas:

- **Snowman:** Add three white spheres, stack them on top of each other. Use small spheres for eyes and buttons, and a cone for the nose.

- **Simple House:** Add a cube for the house body, a pyramid on top for the roof, and small rectangles for the door and windows.

Subtopic 3: Stickers and Text

- **Stickers:** These are ready-made images you can stick onto your 3D models. Click the "Stickers" icon, choose one, and place it on your object.
- **Text:** You can add text. Click the "Text" icon. Choose between **2D text** (flat) or **3D text** (which you can rotate). Type your message and use the tools to change the font and color.

Subtopic 4: Saving and Exporting

- **Saving Your Project:** This saves the file so you can edit it later.
  - Click the `Menu` button (top left) → `Save` or `Save As` → choose a location and file name → click `Save`. It saves as a **.3mf** file.
- **Exporting:** This turns your project into a file you can share, like a picture or video.
  - **As an Image:** `Menu` → `Save as` → `Image` → choose a format like PNG or JPEG.
  - **As a Video:** `Menu` → `Save as` → `Video`. This creates a video that rotates your 3D model.

Evaluation Questions:

1. What is Paint 3D used for?
2. Name three 3D shapes you can make in Paint 3D.
3. How do you rotate a 3D object?
4. What is the main difference between a 2D and a 3D object?
5. How do you save your Paint 3D project so you can edit it later?

WEEK 3: Data Processing

Topic: Data Processing

Subtopic 1: Meaning of Data Processing

- **Data:** These are raw, unprocessed facts and figures. They don't have much meaning on their own. (e.g., the names "Tunde" and "Aisha," or the number "85").
- **Information:** This is data that has been processed, organized, and given meaning. (e.g., "Tunde scored 85 marks in Maths.")
- **Data Processing:** It is the process of collecting raw data and converting it into useful information.

Subtopic 2: Stages of Data Processing

1. **Data Collection (Input):** Gathering the raw data. (e.g., using a form to collect students' names and ages).
2. **Validation:** Checking the data to make sure it is accurate and correct. (e.g., making sure an age isn't written as "200").
3. **Processing:** Performing operations on the data, like sorting, calculating, or grouping. (e.g., arranging names in alphabetical order).
4. **Storage:** Saving the processed information for future use. (e.g., saving student records on a computer).
5. **Retrieval (Output):** Getting the stored information when it's needed. (e.g., a teacher printing a report card).
6. **Communication:** Sharing the information with others. (e.g., sending the report card to parents).

Subtopic 3: Manual vs. Electronic Data Processing

- **Manual Data Processing:**
  - **How:** Done by hand using tools like pen, paper, and files.
  - **Example:** A shopkeeper writing sales in a notebook.
  - **Advantages:** Cheap, simple, doesn't need electricity.

- **Disadvantages:** Very slow, prone to errors, takes up a lot of space, tiring.

- **Electronic Data Processing:**
  - **How:** Done using computers, software, and the internet.
  - **Example:** A school management system that stores all student records.
  - **Advantages:** Very fast, accurate, can handle huge amounts of data, easy to search.
  - **Disadvantages:** Expensive, needs electricity, requires training, can have technical problems.

Feature	Manual Processing	Electronic Processing
Speed	Slow	Fast
Accuracy	Many errors	Few errors
Storage	Takes a lot of physical space	Takes very little space
Cost	Cheap	Expensive

Subtopic 4: Importance in Real Life

- **Education:** For keeping student records, making timetables, and generating report cards.
- **Banking:** For managing accounts, processing ATM transactions, and transferring money.
- **Healthcare:** For storing patient records and booking appointments.
- **Business:** For tracking sales, managing inventory, and keeping customer data.

Evaluation Questions:

1. What is the difference between data and information?
2. What is data processing?
3. List any four stages of data processing.
4. Give one advantage and one disadvantage of electronic data processing.
5. Give one real-life example of data processing in a school.

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WEEK 4: Basic Computer Communication

Topic: Basic Computer Communication

Subtopic 1: Meaning and Importance of Computer Communication

- **What is it?** It is the process of sharing data, information, and resources between two or more computers or devices.
- **Key Elements:**

1. **Sender:** The device starting the communication.
2. **Message:** The information being sent.
3. **Medium:** The path the message travels on (like a cable or Wi-Fi).
4. **Receiver:** The device the message is for.
5. **Protocol:** The set of rules that devices follow to communicate.

- **Importance:**

  - **Fast Information Sharing:** Send messages and files instantly.
  - **Global Reach:** Connect with people all over the world.
  - **Cost-effective:** Sharing one internet connection or printer saves money.
  - **Collaboration:** People can work together on projects from different locations.

Subtopic 2: Types of Computer Communication

- **Wired Communication:** Uses physical cables to connect devices.

  - **Examples:**

    - **Ethernet Cable:** The most common type, used to connect computers to a router.
    - **Fiber Optic Cable:** Uses light to send data very fast over long distances.
  - **Advantages:** Fast, reliable, secure.
  - **Disadvantages:** Limits movement, cables can be messy, installation needed.

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- **Wireless Communication:** Uses electromagnetic waves (like radio signals) to send data without cables.

  - **Examples:**

    - **Wi-Fi:** A wireless network for internet access in a home, school, or café.
    - **Bluetooth:** For short-range connections between devices, like a phone and wireless headphones.
    - **Mobile/Cellular Networks (3G, 4G, 5G):** For internet and calls on your mobile phone.
  - **Advantages:** Allows you to move around, no cables needed.
  - **Disadvantages:** Can be slower than wired, less secure, has a limited range.

Feature	Wired Communication	Wireless Communication
Speed	Faster	Slower
Mobility	Limited	High
Security	More secure	Less secure
Setup	Complex (needs cables)	Easy

Subtopic 3: Introduction to the Internet

- **What is the Internet?** It is a huge, global network of millions of computers all connected together, allowing them to share information.
- **How to Access the Internet:**

1. A device (computer, phone, tablet).
2. An internet connection (like Wi-Fi or mobile data).
3. An account with an **ISP (Internet Service Provider)** like MTN, Glo, or Airtel.
4. A **web browser** (like Chrome or Firefox) to view websites.

- **What You Can Do on the Internet:**

  - Communicate (email, WhatsApp, video calls).
  - Find information (Google search, Wikipedia).

- Learn (online courses, YouTube tutorials).
- Be entertained (Netflix, YouTube, music, games).
- Shop online (Jumia, Konga).
- Do banking.

Evaluation Questions:

1. What is computer communication?
2. Name the five elements of computer communication.
3. Give two examples of wired and two examples of wireless communication.
4. What is the Internet?
5. List three things you can do on the Internet.

WEEK 6: Introduction to Computer Networks

Topic: Introduction to Computer Networks

Subtopic 1: What is a Computer Network?

- **Definition:** A computer network is a group of two or more computers (or other devices) that are connected together to share resources and information.
- **Simple Example:** The computers in your school's computer lab are likely connected in a network.

Subtopic 2: Components of a Network

These are the physical parts and software needed to build a network:

1. **Computers/Devices:** The computers, laptops, tablets, or phones that are connected. They are also called "nodes" or "hosts".
2. **Network Interface Card (NIC):** A piece of hardware inside each computer that allows it to connect to the network.

3. **Cables:** The physical wires (like Ethernet cables) used to connect devices in a wired network.
4. **Switch:** A device that connects multiple devices on the same network, allowing them to communicate with each other.
5. **Router:** A device that directs data traffic between different networks. It connects your home network to the internet.
6. **Modem:** A device that converts signals from your ISP into a signal your devices can use. It's often combined with a router.
7. **Server:** A powerful computer that provides services to other computers on the network, like storing files or managing email.
8. **Network Software:** The operating system and rules (protocols) that make the network work.

Subtopic 3: Types of Networks

- **LAN (Local Area Network):** Covers a small area, like a single building (school, office, home). It is usually fast and cheap to set up.
- **WAN (Wide Area Network):** Covers a large geographical area, like a city, country, or the whole world. The **Internet is the largest WAN**.
- **MAN (Metropolitan Area Network):** Covers an area larger than a LAN but smaller than a WAN, like a city.

Subtopic 4: Benefits of Networking

1. **Resource Sharing:** Multiple computers can share a single printer, an internet connection, or files.
2. **Communication:** Users can easily send messages and files to each other (e.g., email, chat).
3. **Centralized Data Storage:** Files can be stored on a central server, making it easy to back up and access files from any computer on the network.
4. **Cost Savings:** It's cheaper to buy one powerful printer for a network to share than to buy a printer for every single computer.

5. **Collaboration:** People can work together on the same project and share ideas easily.

Evaluation Questions:

1. What is a computer network?
2. Name any four components of a network.
3. What is the difference between a LAN and a WAN?
4. Give two benefits of networking.
5. What is the job of a router?

WEEK 8: Basic Internet Concepts

Topic: Basic Internet Concepts

Subtopic 1: Meaning of the Internet
The internet is a global system of interconnected computer networks that links devices worldwide. It is a "network of networks."

Subtopic 2: Common Internet Terms

1. **Website:** A collection of related web pages, usually under a common name or address (e.g., `www.google.com`).
2. **URL (Uniform Resource Locator):** The unique address of a website or a specific page on the internet (e.g., `https://www.google.com`).
3. **Browser:** A software application used to access and view websites (e.g., Chrome, Firefox, Safari, Edge).
4. **Search Engine:** A website that helps you find information on the internet by searching for keywords (e.g., Google, Bing, Yahoo).
5. **Homepage:** The main or first page of a website.
6. **Hyperlink (Link):** Clickable text or an image that takes you to another web page.

7. **Download:** The process of copying a file from the internet to your own device (computer or phone).
8. **Upload:** The process of sending a file from your own device to the internet (e.g., posting a photo on social media).
9. **Streaming:** Watching or listening to media (video, music) directly from the internet without fully downloading it first (e.g., YouTube, Spotify).
10. **ISP (Internet Service Provider):** A company that provides access to the internet (e.g., MTN, Glo, Airtel in Nigeria).
11. **Email (Electronic Mail):** A method of sending messages and files digitally over the internet.
12. **Social Media:** Websites and apps that allow people to create and share content and connect with others (e.g., Facebook, Instagram, TikTok).
13. **Cloud:** Online storage space on the internet where you can save files and access them from anywhere (e.g., Google Drive, iCloud).

Subtopic 3: Responsible Internet Use

- **Use it for good purposes:** For learning, researching, and positive communication.
- **Manage your time:** Set limits for how long you are online.
- **Respect others:** Be polite, don't say mean things, and respect people's privacy.
- **Verify information:** Not everything you see online is true. Check with other sources.
- **Ask for permission:** Always ask your parents before creating accounts, downloading things, or sharing personal information.

Evaluation Questions:

1. What is the internet?
2. Explain the difference between a browser and a search engine.
3. What is the difference between downloading and uploading?
4. Name any three common internet terms and explain them.
5. List two rules for responsible internet use.

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WEEK 9: Online Safety Rules

Topic: Online Safety Rules

Subtopic 1: Dangers of the Internet

1. **Cyberbullying:** Being bullied online through mean messages, rumors, or threats on social media or in games.
2. **Online Predators:** Strangers who pretend to be friends online to try and trick or harm you.
3. **Inappropriate Content:** Seeing videos, pictures, or information that is violent, disturbing, or not meant for your age.
4. **Scams and Fraud:** Tricks to steal your money or personal information (e.g., fake "You won a prize!" messages).
5. **Identity Theft:** Someone stealing your personal information (like your name and address) to pretend to be you.
6. **Malware (Viruses):** Harmful software that can damage your computer or steal your data.
7. **Oversharing:** Sharing too much personal information online, which can be used against you.
8. **Addiction:** Spending too much time online, which can affect your studies, health, and relationships.

Subtopic 2: Safe Browsing Habits

1. **Keep personal information private.** Never share your full name, address, school, phone number, or passwords online.
2. **Use strong passwords.** Create passwords with at least 8 characters, using a mix of letters (upper and lower case), numbers, and symbols. Don't use your name or birthday.

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3. **Think before you click.** Don't click on suspicious links or download files from people or websites you don't know.
4. **Check website security.** Look for a padlock icon and `https://` in the address bar before entering any personal information on a site.
5. **Use privacy settings.** Make your social media accounts private so only people you know can see your posts.
6. **Don't meet online friends.** People you meet online may not be who they say they are. Never agree to meet them without a parent.
7. **Log out.** Always log out of your accounts, especially on shared computers.
8. **Report problems.** Tell a parent or teacher immediately if someone is mean to you, asks for personal information, or makes you feel uncomfortable online.

Subtopic 3: Responsible Use of Digital Devices

- **Set screen time limits.** Take breaks and don't spend too much time on your devices.
- **Balance your activities.** Make sure you have time for homework, physical activity, and family, not just screen time.
- **Take care of your device.** Handle it gently, keep it away from water, and charge it properly.
- **Protect your posture.** Sit up straight and keep your device at eye level.
- **Follow family rules.** Obey your parents' rules about when and where you can use your devices.

Evaluation Questions:

1. Name three dangers of the internet.
2. List three pieces of personal information you should never share online.
3. What is cyberbullying, and what should you do if it happens to you?
4. Give two examples of safe browsing habits.
5. Why is it important to have a strong password?

WEEK 10: Online Privacy and Security

Topic: Online Privacy and Security

Subtopic 1: Meaning of Online Privacy

Online privacy is about protecting your personal information and controlling who can see it when you are on the internet. It's important to prevent identity theft, scams, and to stay safe from predators.

Subtopic 2: Creating Strong Passwords

A strong password is your first line of defense.

- **A Strong Password MUST have:**
  - At least 8 characters.
  - A mix of uppercase and lowercase letters.
  - Numbers.
  - Special symbols (like `@`, `#`, `$`, `!`).
  - It should NOT be a real word or easy-to-guess information (like your name or birthday).
- **Tips for creating strong passwords:**
  - Use a phrase you can remember: "I love playing football!" could become `ILoveFootball!` or `ILpF@b!!`.
  - Use a different password for each of your important accounts (like email and social media).
  - Never share your passwords with anyone except your parents.

Subtopic 3: Two-Factor Authentication (2FA)

2FA is an extra layer of security. When you log in, you not only enter your password but also a special code that is sent to your phone. This makes it much harder for someone else to break into your account.

Subtopic 4: Avoiding Scams and Phishing

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- **Scams:** Tricks designed to steal your money or information. They often promise something "too good to be true," like a free phone or a prize you never entered.
- **Phishing:** A common scam where you receive a fake email or message that looks like it's from a real company (like your bank). It asks you to click a link and "verify" your information, which then gets stolen.
- **How to Avoid Scams:**
  - Be suspicious of amazing offers.
  - Don't click links in emails from senders you don't know.
  - Real companies will never ask for your password via email.
  - Check for poor spelling and grammar in messages.
  - If you're unsure, show a parent or teacher.

Evaluation Questions:

1. What is online privacy?
2. List three rules for creating a strong password.
3. What is two-factor authentication (2FA)?
4. What is a phishing scam?
5. Name two warning signs of an online scam.

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WEEK 11: Revision

Topic: Comprehensive Revision of Second Term

This week is for reviewing all the topics we have covered this term. Go through your notes and practice answering the evaluation questions. Ask your teacher if there's anything you don't understand.

Key Topics to Review:

1. **MS PowerPoint:** Creating slides, adding text/images, transitions vs. animations, running a slideshow.
2. **Paint 3D:** 2D vs. 3D, creating basic 3D shapes, rotating and coloring objects, saving/exporting.
3. **Data Processing:** Data vs. Information, the six stages, manual vs. electronic processing.
4. **Computer Communication:** Wired vs. wireless communication, the internet, its uses.
5. **Computer Networks:** Definition, components (router, switch, etc.), LAN vs. WAN, benefits.
6. **Internet Concepts:** Key terms like URL, browser, search engine, download, upload.
7. **Online Safety:** Dangers (cyberbullying, predators), safe browsing habits, responsible device use.
8. **Online Privacy & Security:** Strong passwords, 2FA, avoiding scams and phishing.

WEEK 12: Examination

Topic: Second Term Examination

Get a good night's rest and do your best on your exam. Read each question carefully before you answer.

WEEK 13: Examination

Topic: Second Term Examination (Continued)

WEEK 14: Marking and Review of Scripts

Topic: Marking and Review of Scripts

Your teacher will mark your exams. This week, you will go over the exam questions and answers to learn from your mistakes and understand the topics better.

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